

Home Assignment - Chunking Playground - Document Chunking

Assignment 1: Document Chunking

1. Fixed-Size Character Chunking

The screenshot displays the 'Chunking Playground' web application interface, which is used for visualizing RAG document segmentation. The interface is divided into several sections:

- Left Sidebar:** Contains options for 'Upload File', 'Web URL', 'Sample Web', 'Upload Document', 'Sample Library' (with items like 'HR Leave Policy', 'Company FAQ', 'RAG Research Abstract', 'Banking Loan Policy'), 'Chunking Strategy' (set to 'Fixed-Size Character Chunking'), and 'Fixed-Size Character Chunking'.
- Document Preview:** Displays the text of the 'EMPLOYEE LEAVE POLICY AND GUIDELINES' document. It includes sections for 'Purpose and Scope', 'Annual Paid Time Off (PTO)', 'Sick Leave', and 'Parental Leave'. Below the text, statistics are shown: 'H: 270 words', 'A: 1295 characters', 'S: 44 sentences', 'P: 5 paragraphs', and 'T: 324 tokens'.
- Analysis/Segmentation:** Shows 'TOTAL CHUNKS: 3', 'AVG. CHUNK SIZE: 465 chars', 'EST. TOKENS: 349', and 'TIME ELAPSED: 0.506 ms'.
- Teaching Mode:** Explains 'Fixed-Size Character Chunking' as a straightforward strategy that splits text into chunks of exact character lengths with a specific overlap. It lists 'Advantages' (Extremely fast, guarantees predictability, no dependencies) and 'Disadvantages' (Ignores natural language structures, often breaks sentences, can cause context loss). It also notes 'Why Companies Use It' (Very easy to implement).
- Chunking Strategy Configuration:** A section on the left allows setting 'Chunk Size (chars)' to 500 and 'Overlap (chars)' to 50. It includes an 'Apply Chunking' button and a 'Ingest into MongoDB' button.
- Chunks View:** A section at the bottom titled 'FIXED-SIZE CHUNKING EDUCATIONAL INSIGHTS' includes 'Boundary Violations & Risk' (Sentence boundaries broken: 2 times, Word boundaries broken: 1 time, Context Loss Risk: High) and a 'Would Semantic Chunking perform better?' section explaining why semantic chunking is superior.
- Chunk Timeline & Overlap Visualizer:** A horizontal bar chart showing the document's structure and the placement of three chunks (Chunk #1, Chunk #2, Chunk #3) with their respective overlaps.

Chunk #1

500 chars

Characters: 0-500

Start: 0 End: 500 Overlap: 0 chars

Sentence Boundary Broken

EMPLOYEE LEAVE POLICY AND GUIDELINES 1. Purpose and Scope This policy outlines the guidelines and procedures governing various types of leave available to all full-time employees. The company believes that balance between professional and personal life is essential for productivity and well-being. 2. Annual Paid Time Off (PTO) All full-time employees accumulate PTO at a rate of 1.67 days per month, resulting in a total of 20 days per calendar year. - PTO requests must be submitted at least 5

500 ch

80 w

6 sent

3 para

125 tok

Chunk #2

500 chars

Characters: 450-950

Start: 450 End: 950 Overlap: 50 chars

Word Boundary Broken

Sentence Boundary Broken

Overlap Region

OVERLAP START

our

- PTO requests must be submitted at least 5

OVERLAP END

our. - PTO requests must be submitted at least 5 business days in advance for approval by the department manager. - A maximum of 5 unused PTO days may be carried over to the next calendar year. Any additional unused days will expire. 3. Sick Leave Employees receive 10 paid sick days per year on January 1st. Sick leave is intended for personal illness, medical appointments, or caring for immediate family members. - If sick leave exceeds 3 consecutive days, a certified medical note from a qualif

500 ch

87 w

8 sent

3 para

125 tok

Chunk #3

395 chars

Characters: 900-1295

Start: 900 End: 1295 Overlap: 50 chars

Overlap Region

OVERLAP START

utive days, a certified medical note from a qualif

OVERLAP END

utive days, a certified medical note from a qualified physician is required. 4. Parental Leave The company provides up to 12 weeks of paid parental leave to eligible employees following the birth, adoption, or foster placement of a child. Employees must have completed at least one year of continuous service to qualify. - Parental leave runs concurrently with FMLA guidelines where applicable.

395 ch

62 w

5 sent

2 para

99 tok

2. Recursive Character Chunking

Chunking Strategy

Recursive Character Chunking

Chunk Size (chars)500

Overlap (chars)50

Chunks ViewComparison DashboardRAG Simulator

Proceed to Ingest into MongoDBTree ViewGrid View

Recursive Character Chunking Educational Insights

Hierarchy Preservation Metrics

Paragraph Preservation %:100%

Line Preservation %:100%

Word Breaks Count:0 breaks

Character Splits (Forced):0 splits

Separator Usage Analytics

Paragraph (%n%)3

Line (%n)0

Word (%)0

Char (%)0

Fixed vs Recursive Split Comparison

Metric	Fixed Chunking	Recursive Chunking
Paragraph Preservation	68%	100%
Line Preservation	82.4%	100%
Word Boundary Breaks	1	0
Sentence Boundary Breaks	2	0

Dynamic Hierarchy Preservation

Why is Recursive better than Fixed?

Fixed-size chunking slices text strictly by character length, which frequently splits words (e.g., comput-er) and cuts sentences in half. Recursive chunking uses a hierarchical fallback system: it attempts splits on paragraphs first, then lines, then word spaces. This ensures word boundaries are never broken and sentences remain complete inside chunks.

Why is Recursive STILL not Semantic?

While recursive chunking preserves structural text segments (like paragraphs and lines), it does not analyze the semantic meaning or content transitions of sentences. If a single topic spans across paragraphs, recursive splitting might partition them into separate chunks purely based on size limits, whereas Semantic Chunking evaluates topic embedding similarities to group related paragraphs coherently.

Recursive Process Walking Simulator

Show Recursive Process

Chunk #1299 chars

Split Separator Used: \n\n

Split Reason: Chunk size reached of paragraph boundary. (Level: Paragraph)

EXCELLENT SPLIT

EMPLOYEE LEAVE POLICY AND GUIDELINES

1. Purpose and Scope This policy outlines the guidelines and procedures governing various types of leave available to all full-time employees. The company believes that balance between professional and personal life is essential for productivity and well-being.

299 ch42 w3 sent2 para75 tok

Chunk #2383 chars

Split Separator Used: \n\n

Split Reason: Chunk size reached of paragraph boundary. (Level: Paragraph)

EXCELLENT SPLIT

2. Annual Paid Time Off (PTO) All full-time employees accumulate PTO at a rate of 1.67 days per month, resulting in a total of 20 days per calendar year. - PTO requests must be submitted at least 5 business days in advance for approval by the department manager. - A maximum of 5 unused PTO days may be carried over to the next calendar year. Any additional unused days will expire.

383 ch71 w5 sent1 para96 tok

Chunk #3290 chars

Split Separator Used: \n\n

Split Reason: Chunk size reached of paragraph boundary. (Level: Paragraph)

EXCELLENT SPLIT

3. Sick Leave Employees receive 10 paid sick days per year on January 1st. Sick leave is intended for personal illness, medical appointments, or caring for immediate family members. - If sick leave exceeds 3 consecutive days, a certified medical note from a qualified physician is required.

290 ch47 w4 sent1 para72 tok

Chunk #4317 chars

Split Separator Used: None

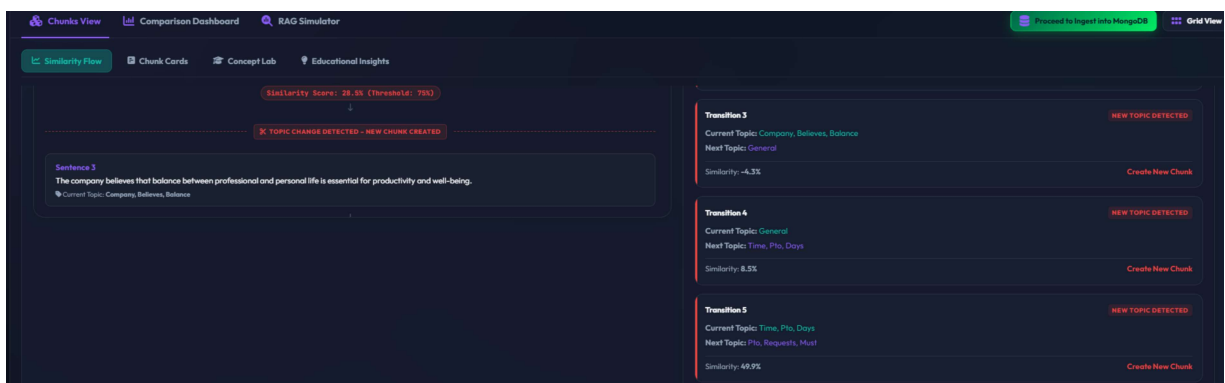
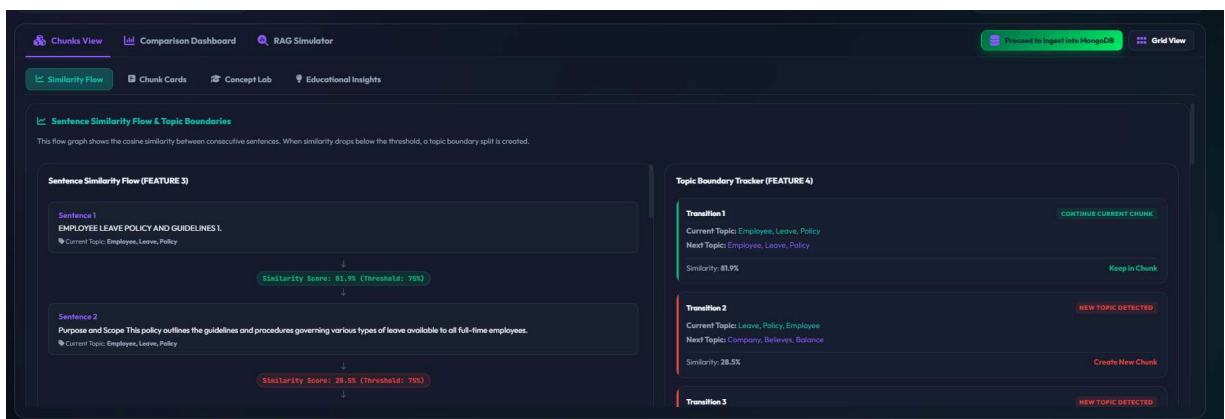
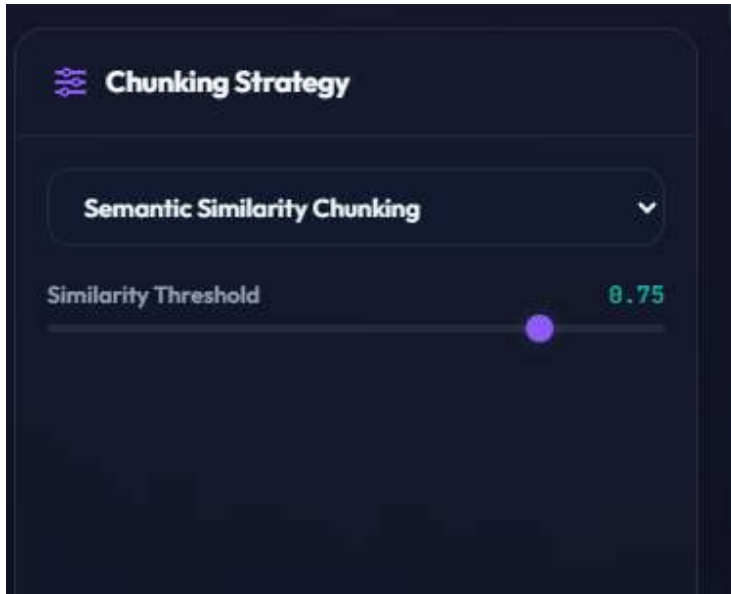
Split Reason: Reached the end of the document. (Level: Document End)

EXCELLENT SPLIT

4. Parental Leave The company provides up to 12 weeks of paid parental leave to eligible employees following the birth, adoption, or foster placement of a child. Employees must have completed at least one year of continuous service to qualify. - Parental leave runs concurrently with FMLA guidelines where applicable.

317 ch50 w4 sent1 para79 tok

3. Semantic Similarity Chunking



	Similarity: 44.9%	Create New Chunk
Transition 6	NEW TOPIC DETECTED	
Current Topic: Pto, Requests, Must		
Next Topic: Maximum, Unused, Pto		
Similarity: 54%	Create New Chunk	
Transition 7	NEW TOPIC DETECTED	
Current Topic: Maximum, Unused, Pto		
Next Topic: Additional, Unused, Days		
Similarity: 59.6%	Create New Chunk	
Transition 8	NEW TOPIC DETECTED	
Current Topic: Additional, Unused, Days		
Next Topic: General		
Similarity: 4.3%	Create New Chunk	

Transition 9	NEW TOPIC DETECTED	
Current Topic: General		
Next Topic: Sick, Leave, Employees		
Similarity: 8.9%	Create New Chunk	
Transition 10	NEW TOPIC DETECTED	
Current Topic: Sick, Leave, Employees		
Next Topic: Sick, Leave, Intended		
Similarity: 61.5%	Create New Chunk	
Transition 11	NEW TOPIC DETECTED	
Current Topic: Sick, Leave, Intended		
Next Topic: Sick, Leave, Exceeds		
Similarity: 65.2%	Create New Chunk	

Transition 12	NEW TOPIC DETECTED	
Current Topic: Sick, Leave, Exceeds		
Next Topic: General		
Similarity: 0.2%	Create New Chunk	
Transition 13	NEW TOPIC DETECTED	
Current Topic: General		
Next Topic: Parental, Leave, Company		
Similarity: -1.1%	Create New Chunk	
Transition 14	NEW TOPIC DETECTED	
Current Topic: Parental, Leave, Company		
Next Topic: Employees, Must, Completed		

Transition 15

NEW TOPIC DETECTED

Current Topic: Employees, Must, Completed

Next Topic: Parental, Leave, Runs

Similarity: 44.7%

Create New Chunk

4. Metadata-Enhanced

Chunking Strategy

Metadata-Enhanced

Chunk Size (chars)100

Overlap (chars)50

Metadata Settings

Chunks ViewComparison DashboardRAG Simulator

Proceed to Ingest into MongoDBGrid View

Chunk CardsTimeline FlowStrategy Comparison

Search metadata values...All DepartmentsClear Filters

Chunk #1Topic: Leave / Policy / Guidelines

T18: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRange: GlobalSecu: InternalVer: 1.0Word: 210Enti: 323Targ: Standard Users38 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Introduction]

EMPLOYEE LEAVE POLICY AND GUIDELINES

Chunk #2Topic: Outlines / Guidelines / Policy

T18: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRange: GlobalSecu: InternalVer: 1.0Word: 210Enti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Purpose and Scope]

1. Purpose and Scope This policy outlines the guidelines and procedures governing various types of l

Chunk #3Topic: Available / Employees / Elmes

T18: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRange: GlobalSecu: InternalVer: 1.0Word: 210Enti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Purpose and Scope]

elines and procedures governing various types of leave available to all full-time employees. The com

Chunk #4Topic: Balance / Professional / Employees

T18: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRange: GlobalSecu: InternalVer: 1.0Word: 210Enti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Purpose and Scope]

eeve available to all full-time employees. The company believes that balance between professional an

Chunk #5Topic: Balance / Pany / Professional

T18: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRange: GlobalSecu: InternalVer: 1.0Word: 210Enti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Purpose and Scope]

pany believes that balance between professional and personal life is essential for productivity and

Chunk #6

Topic: Being / Personal / Productivity

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users65 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Purpose and Scope]

d personal life is essential for productivity and well-being.

Chunk #7

Topic: Time / Employees / Annual

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

2. Annual Paid Time Off (PTO) All full-time employees accumulate PTO at a rate of 1.67 days per mont

Chunk #8

Topic: Days / Rate / Resulting

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

ees accumulate PTO at a rate of 1.67 days per month, resulting in a total of 20 days per calendar ye

Chunk #9

Topic: Submitted / Must / Least

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

h, resulting in a total of 20 days per calendar year. - PTO requests must be submitted at least 5 b

Chunk #10

Topic: Submitted / Must / Approval

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

ar. - PTO requests must be submitted at least 5 business days in advance for approval by the depart

Chunk #11

Topic: Days / Manager / Unused

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

usiness days in advance for approval by the department manager. - A maxiaum of 5 unused PTO days may

Chunk #12

Topic: Manager / Unused / Over

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

ment manager. - A maximum of 5 unused PTO days may be carried over to the next calendar year. Any ad

Chunk #13

Topic: Additional / Unused / Over

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users85 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Annual Paid Time Off (PTO)]

be carried over to the next calendar year. Any additional unused days will expire.

Chunk #14

Topic: Sick / Leave / Receive

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Sick Leave]

3. Sick Leave Employees receive 10 paid sick days per year on January 1st. Sick leave is intended fo

Chunk #15

Topic: Sick / Personal / Illness

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Sick Leave]

per year on January 1st. Sick leave is intended for personal illness, medical appointments, or carin

Chunk #16

Topic: Sick / Personal / Illness

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Sick Leave]

r personal illness, medical appointments, or caring for immediate family members. - If sick leave ex

Chunk #17

Topic: Sick / Consecutive / Note

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Sick Leave]

g for immediate family members. - If sick leave exceeds 3 consecutive days, a certified medical note

Chunk #18

Topic: Consecutive / Required / Cards

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users92 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Sick Leave]

ceeds 3 consecutive days, a certified medical note from a qualified physician is required.

Chunk #19

Topic: Leave / Parental / Employee

T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Parental Leave]

4. Parental Leave The company provides up to 12 weeks of paid parental leave to eligible employees f

Chunk #20

Topic: Birth / Employees / Adoption

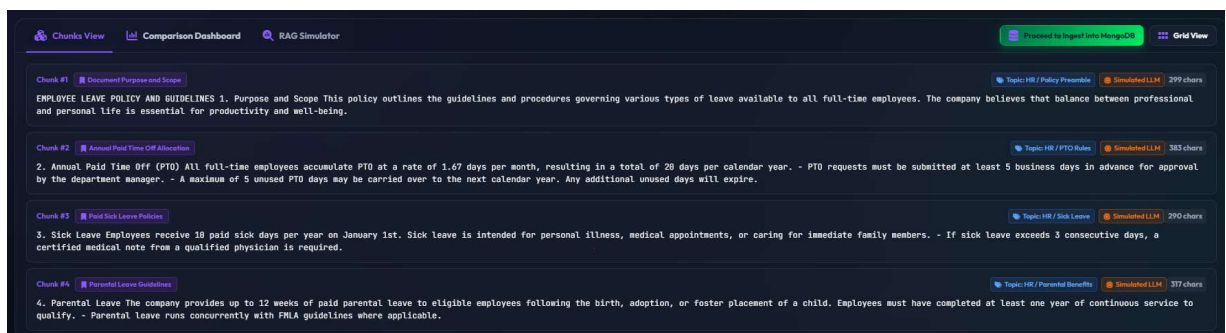
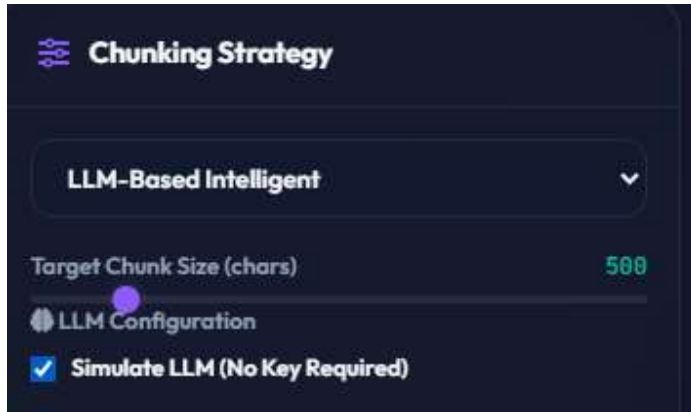
T1B: Medical Diagnosis ChartDoc: HealthcareLang: EnglishDepth: Clinical OperationsRegi: GlobalSecu: InternalVer: 1.0Word: 210Esti: 323Targ: Standard Users100 chars

Context Prepend: [Dept: Clinical Operations | Ver: 1.0 | Section: Parental Leave]

eks of paid parental leave to eligible employees following the birth, adoption, or foster placement

Load More Chunks (20 of 24 shown)

5. LLM-Based Intelligent





MongoDB Ingestion

Partitioned Document & Chunk Storage





Ingestion & Embedding Complete

Each chunk has been stored as an individual MongoDB document with a 1,024-dimensional Mistral embedding.

TARGET COLLECTION

document_chunks

STORAGE ARCHITECTURE

 Individual Chunk Documents

EMBEDDING MODEL

 Mistral (1,024 Dimensions)

VECTOR SEARCH STATUS

 Ready (vector_index)

DOCUMENT NAME

HR Leave Policy

DOCUMENTS INGESTED

4 Documents

FIRST INSERTED ID

6ab8008f013d2853743469dd

 **Verify Vector Search**

1,024-d Cosine Similarity

 **Search**

Close

document_chunks

Individual Chunk Documents

● Mistral (1,024 Dimensions)

🟢 Ready (vector_index)

HR Leave Policy

4 Documents

6ab8008f013d2853743469dd

1,024-d Cosine Similarity

▶ Search

Close

ORGANIZATION
vigneshwaran's Org

PROJECT
Project 0

Compare tiers

My Queries

Data Modeling

CLUSTERS (0)

Search clusters

Cluster0

logtest_cases

Resume_RAG

admin

chunking_playground

document_chunks

resume_chunks

web_url_chunks

local

document_chunks

Cluster0 > chunking_playground > document_chunks

Documents 4

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 4 of 4

_id: ObjectId("6ab8888f613d2853743468dd")

document_id: "fdd9f766-bbe2-42c5-97f5-7a9c987c99b4"

document_name: "HR Leave Policy"

source_type: "document"

source_url: null

strategy: "llm"

strategy_params: Object (2)

chunk_id: 1

chunk_index: 1

total_chunks: 4

text: "EMPLOYEE LEAVE POLICY AND GUIDELINES"

1. Purpose and Scope,"

char_count: 299

start_char: null

end_char: null

overlap_size: 0

metadata: Object (10)

embedding: Array (1024)

embedding_model: "mistral-embed"

embedding_dimensions: 1024

ingested_at: "2020-09-20T17:27:43.119817+06:00"

_id: ObjectId("6ab8888f613d2853743469de")

document_id: "fdd9f766-bbe2-42c5-97f5-7a9c987c99b4"

document_name: "HR Leave Policy"

source_type: "document"

Alert System Status: All Good

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